WaferEngine · CS-3

WEEKLY COLLABORATOR DISCUSSION

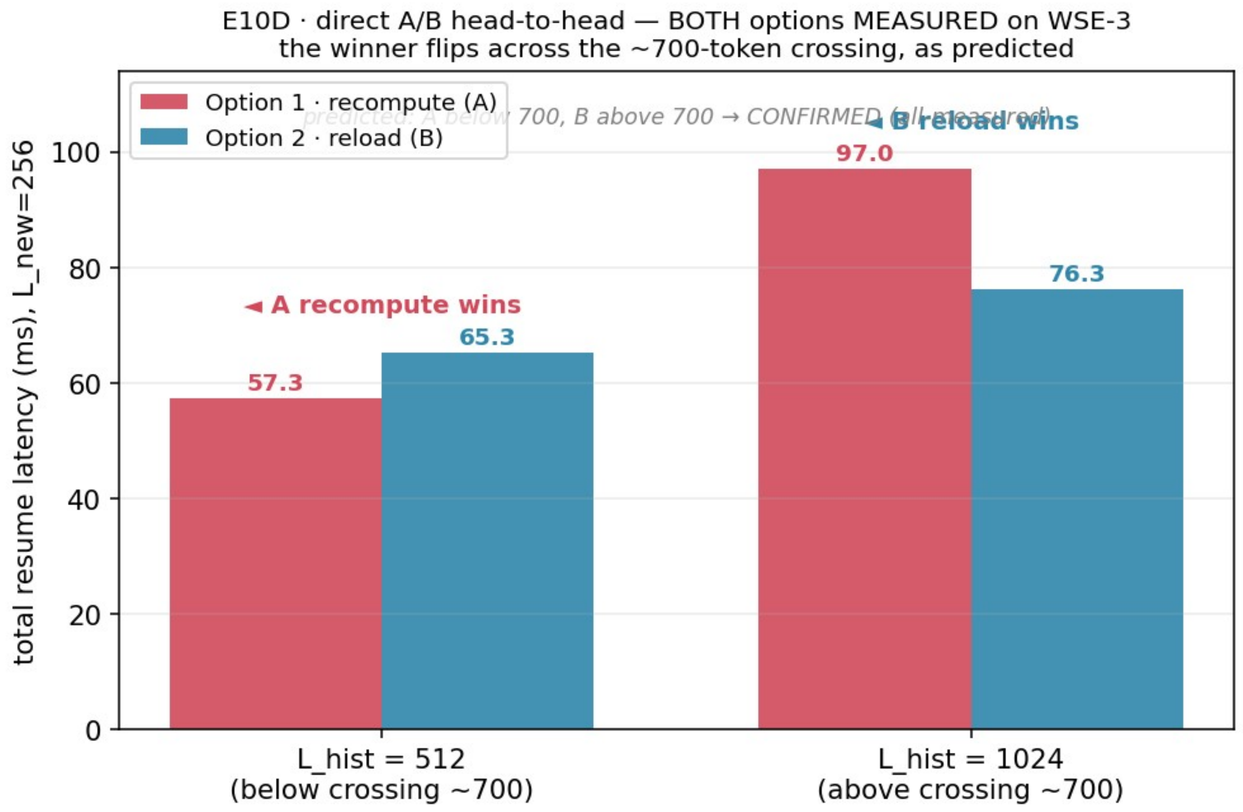
On-Chip KV Offload as a Storage Tier

Two multi-row storage designs measured on real WSE-3; the on-chip tier now sits under both prior resume options.

MODEL	EVIDENCE	SCOPE	DATE
Qwen3-1.7B geometry	81 CS-3 runs, 0-cycle spread	M3 idle-PE tier	2026-08-24

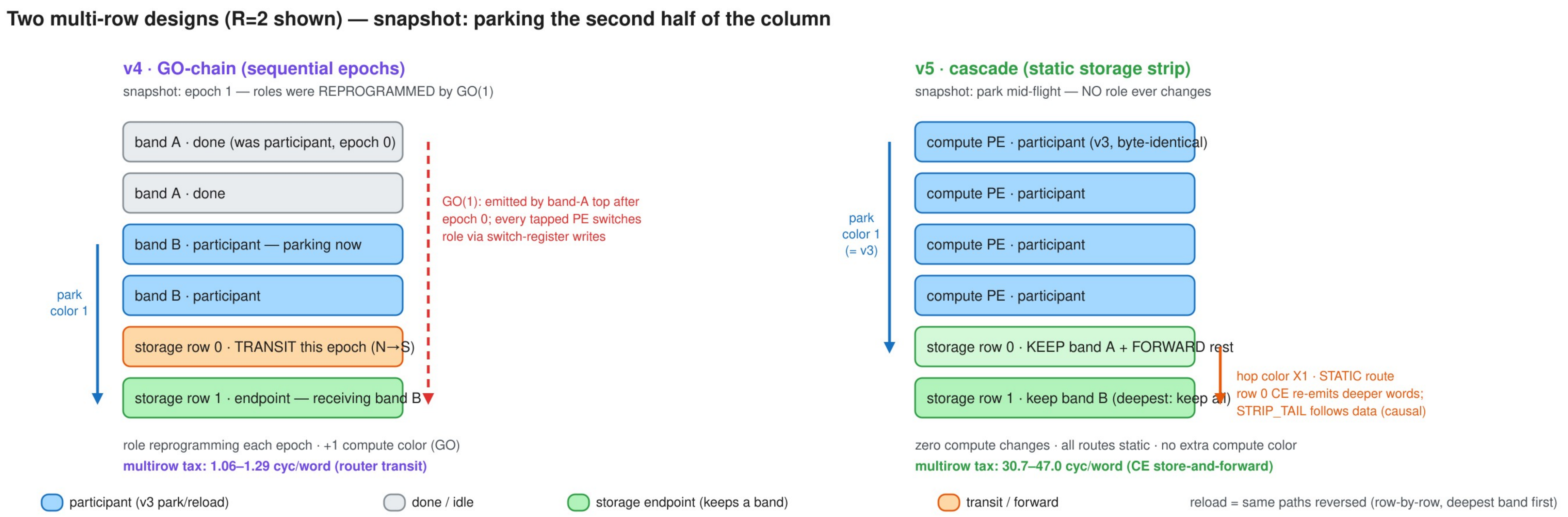
GOAL · WHERE WE ARE IN THE MILESTONES

M2 ended at recompute-vs-host with a 744-token crossing; M3 asks if idle PEs beat both



Last time (M2 E10/E10D, all device-measured): below H*=744 recompute wins, above it host reload wins — but reload rides shared edge IO and the host. M3's question: can idle on-chip PEs hold parked KV as a middle tier? This week: build the multi-row storage strip both candidate ways, measure both end-to-end, and price the tier.

v4 reprograms compute roles per epoch; v5 keeps compute byte-identical and cascades inside the strip



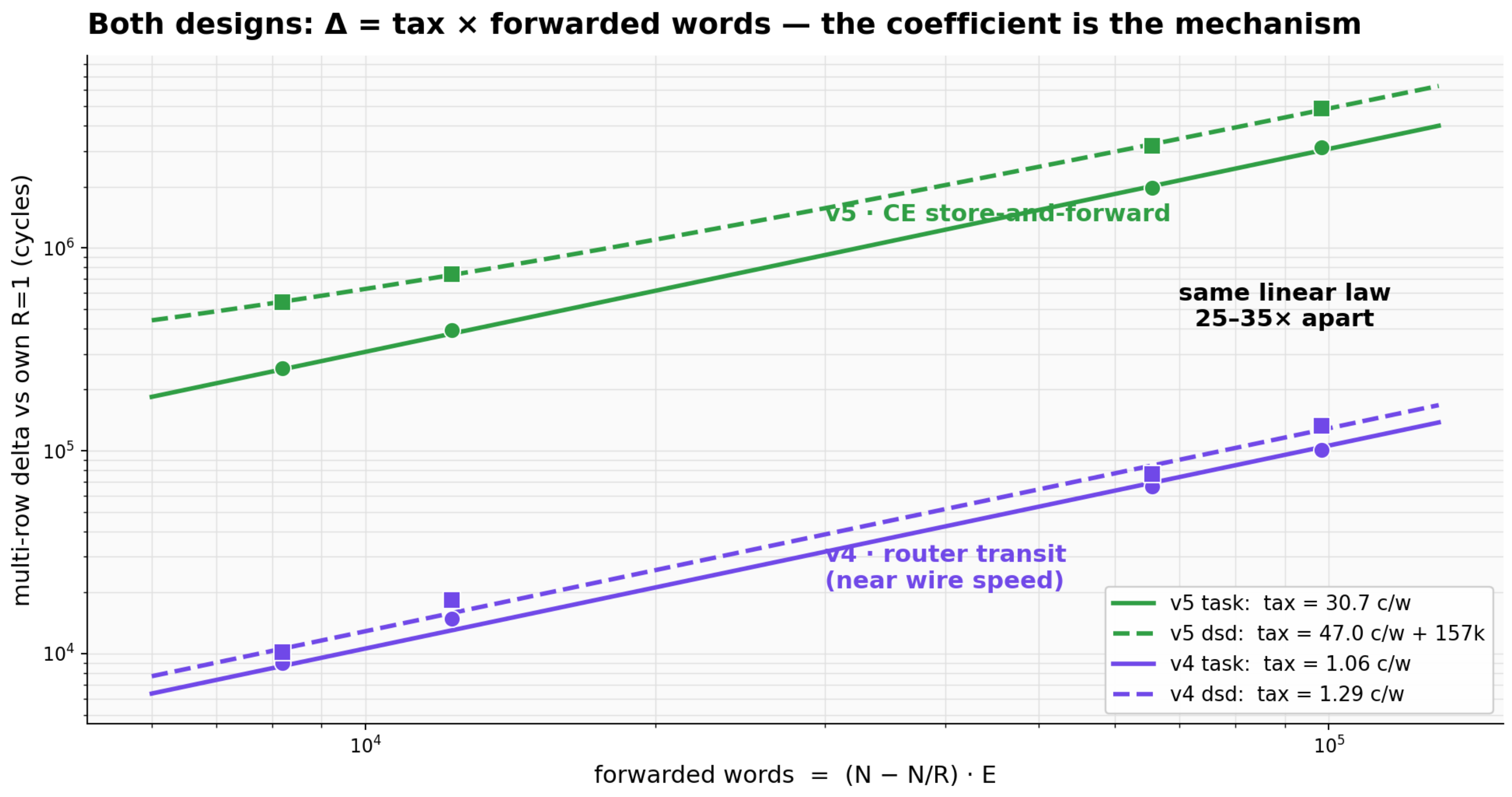
Both share the proven single-row protocol on one borrowed color. v4: bands park in sequential epochs, chained fully on-chip by a GO color; every tapped PE rewrites its switch positions per epoch. v5: row 0 keeps the nearest band and store-and-forwards deeper words on static per-hop colors; a STRIP_TAIL word rides behind the data as the causal end-of-stream. Reload returns over the same paths reversed.

RESULTS · 81 RUNS, EVERY CELL N=3 WITH 0-CYCLE SPREAD

Full park+reload cycle (cycles): v5 wins single-row, v4 wins every multi-row cell

CONFIG (N=256)	V3 SINGLE-ROW	V4 GO-CHAIN	V5 CASCADE
task E=64 · R=1	1,388,470	1,535,920	1,420,244
task E=64 · R=2	—	1,544,946	1,675,327
task E=64 · R=4	—	1,550,855	1,812,526
dsd E=512 · R=1	7,350,234	8,542,434	7,482,812
dsd E=512 · R=2	—	8,619,349	10,669,498
dsd E=512 · R=4	—	8,675,026	12,343,610

Both designs obey delta = tax x forwarded words — the tax is the mechanism, 25-35x apart



Delta vs each design's own R=1 baseline collapses on the same axis, fwd_words = (N - N/R)·E. v4's tax is router-level transit (1.06/1.29 cyc/word, near wire speed); v5's is a CE data-task per forwarded word (30.7/47.0). The v5 dsd fit needs one extra constant (~157k cycles): strip rows emit their own block before relaying, which fast owners expose.

MODEL · THE COEFFICIENT LADDER (ALL DEVICE-FIT)

Every per-word cost in the tier, pinned

39 anchors, <1%

three device-fit models in one file: v3 0.89% / v5 0.83% / v4 0.23% max error

75.4
c_fwd_word cyc/word: row-0 park receive + forward (receive-only was 43.3)

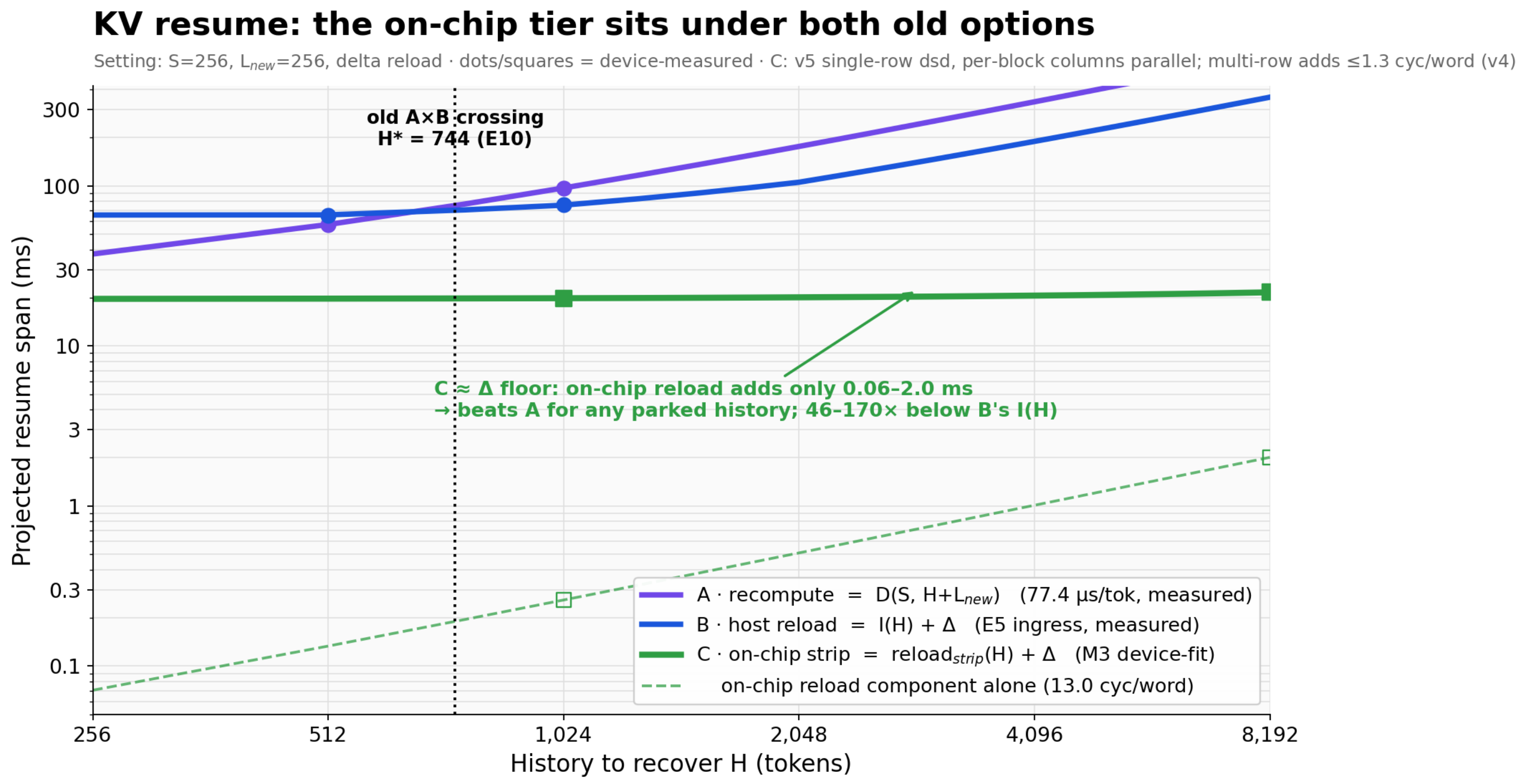
28.8
c_relay_word cyc/word: row-0 reload relay (bare emit loop: 13.0)

1.1 vs 31
multirow tax cyc/word, task: v4 router transit vs v5 CE store-and-forward

+9.6 / +1.6
always-on R=1 premium over v3, cyc/word: v4 roles / v5 cascade branches

SO WHAT · THE TIER HOLDS

The on-chip tier sits under both old options: resume is now delta-dominated



Same axes and setting as the M2 boundary figure, plus measured tier C: on-chip strip reload adds only 0.06-2.0 ms (13.0 cyc/word), 46-170x below host ingress I(H), flattening resume onto the ~20 ms L_new delta floor. The old 744-token crossing becomes a fallback choice for un-parked history. Next lever for resume latency is the delta/restart path, not the transfer.